A

Project Report

on

Nyaya AI

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(Formerly UPTU)

May, 2026

DECLARATION

We hereby declare that the project report titled **"Nyaya.AI – Democratizing Legal Knowledge and Services through AI"** submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (AI & ML) is an original work carried out by us under the guidance of our supervisor.

To the best of our knowledge and belief, this report does not contain any material previously published or written by another person, nor has it been submitted for the award of any other degree or diploma. Wherever references have been made to the work of others, proper acknowledgment has been given.

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CERTIFICATE

This is to certify that the project entitled **"Nyaya.AI – Democratizing Legal Knowledge and Services through AI"** submitted by Sameer Singh Rawat, Shubham Verma, Pooja Chauhan, and Devanshu Sharma in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (AI & ML) from Dr. A.P.J. Abdul Kalam Technical University, Lucknow, is a record of their original work carried out under my supervision.

The work embodied in this report has not been submitted elsewhere for the award of any other degree or diploma.

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ABSTRACT

Many people in India, especially those who live in villages and small towns, still find it hard to reach legal information. Laws plus constitutional rights are in place to protect every citizen but large numbers of Indians cannot grasp or use those rights. The reasons include dense legal wording, absence of awareness, shortage of money and scarce contact with lawyers.

This initiative introduces Nyaya.AI, a web platform that uses artificial intelligence to lower legal complexity but also to deliver answers in multiple regional languages. The service combines three tools - a chatbot that speaks many languages, software that drafts documents on its own and a module that links users to verified lawyers. A person may type a legal question in the language he or she knows best as well as the system returns a plain explanation together with the matching legal source.

The system uses natural language processing methods together with a rule based validation engine. This combination checks that text is clear, fits the context and follows correct procedures. Tests show that users handle the tool well, that the answers it gives are acceptable plus that the time needed to prepare documents drops markedly.

Nyaya.AI seeks to spread legal knowledge, to lower the hurdles that keep people from legal help and to build a justice system that welcomes everyone but also that relies on modern technology.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
NLP	Natural Language Processing
UI	User Interface
API	Application Programming Interface
DB	Database

CHAPTER 1
INTRODUCTION

1.1 Introduction

India's courts and laws exist to give every person fair treatment, equal standing plus the basic freedoms promised by the Constitution. In everyday life, however, large numbers of citizens still reach legal advice or court help only with difficulty. Villagers and people in small towns feel this gap most - they seldom know what the law allows but also they struggle to start a case or defend a claim.

The obstacles are straightforward - statutes and court forms use technical words. Districts speak dozens of languages but rules appear mostly in English. Lawyers ask for fees that poor households cannot pay as well as the nearest legal help office often sits in a distant city.

State and private programmes do offer help but most demand that the person travel to an office, wait for a fixed appointment or bring a middle man. Printed booklets, websites or helplines also operate mainly in English - anyone who reads only Hindi, Tamil, Bengali or another regional tongue must rely on summaries given by others. Because of those steps, common grievances about faulty goods, unpaid wages, family disputes or land boundaries linger for years or fade away without a hearing.

Artificial Intelligence besides Natural Language Processing evolve fast and this progress offers a way to close the accessibility gap. Systems that rely on AI read plain questions, search for matching legal data plus return organised answers within seconds. If such systems also speak many languages and automate documents, the task of getting simple legal direction becomes far easier.

This project starts from the wish to share legal knowledge widely but also to lower the walls that keep people away from it. The goal is not to push aside qualified lawyers but to give a first layer of advice so that each person gains facts for sound choices. Through AI deployed on a website, legal help arrives at once, in clear words and at low cost.

1.2 Objectives of the Project

The project focuses on five main goals:

- It aims to build an AI system that understands and answers questions in multiple languages about legal matters.
- The system will translate complex legal rules and steps into plain language that any user understands.
- The system will draft standard legal papers like rent agreements or power-of-attorney forms without manual effort.
- The platform will include a module that connects users to qualified lawyers whose credentials the project team has checked.
- The entire service will remain easy to reach, simple to navigate plus quick to reply so that legal help becomes more practical for everyone.

1.3 Organization of the Report

This report has five chapters - chapter 1 states the problem area and the goals of the project. Chapter 2 surveys published studies plus describes comparable systems. Chapter 3 sets out the chosen method and the structure of the new system. Chapter 4 gives the test outcomes but also examines what they mean. Chapter 5 ends the report and lists possible next steps for improvement.

1.4 Project Description

Nyaya.AI operates as an online platform that delivers legal help. It uses Artificial Intelligence to place legal knowledge and first step legal services within reach of any user. One interface contains a chatbot that speaks multiple languages, a module that drafts documents, short legal lessons plus a route to book a lawyer.

The chatbot forms the heart of the platform - it accepts questions in English, Hindi, Marathi and further regional languages. Natural language processing reads the question, matches it against a checked legal data set but also returns a clear answer. A rule based layer then reviews the answer so that the procedure and wording stay consistent.

Beside the question answer function, the platform offers document automation. The user picks a document type - complaint, affidavit, legal notice or similar. A short form asks for the necessary facts. Once the form is complete, the system arranges the data into a formatted document as well as offers the file for immediate download. This process shortens drafting time and cuts basic errors in routine paperwork.

A further core element is the “Know Your Rights” area - it offers brief, plain language summaries of key constitutional clauses, statutory entitlements and available legal remedies. All material is sorted under clear legal headings like consumer law, labour rights, property disputes and family law - a user moves straight to the topic that matches the problem.

The platform hosts a lawyer directory - every listed lawyer completed verification. If a user needs full representation, the directory provides direct contact to a licensed lawyer. The site offers initial guidance and transfers the user to a qualified lawyer when the issue lies outside the automated range.
Developers created the front end with React besides Next.js - the backend operates on cloud infrastructure. It manages secure authentication and synchronizes the database in

real time. The artificial intelligence layer replies within seconds, retains context across questions plus changes language without further configuration.

Nyaya.AI positions itself as a link between citizens and the formal legal order - blending ease of access, automated support but also smart help inside a digital structure that expands as demand grows.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to Legal Technology

AI has become a major focus in law during the last ten years - surden [1] explains that machine learning methods change how lawyers study cases - they now automate tasks like document review, prediction of results and research. Susskind [2] adds that digital change alters courts plus the way legal help reaches clients across countries.

People call software built for lawyers “Legal Tech” - such programs aim to speed up, open up and organise legal work through electronic tools. Systems that rely on AI check contracts, run due diligence reviews, help draft briefs but also forecast verdicts. Bennett and Solow Niederman [5] warn that fitting those systems to each country's rules stays hard because statutes and procedures differ.

Those systems work well in wealthy countries but India uses them only in small pockets. Agarwal and Singh [3] list the main barriers - many tongues, court rules that change from one state to the next and users who seldom handle digital tools.

2.2 Existing Legal Assistance Platforms

Online platforms deliver legal support services - their range, ease of access and degree of automation vary widely.

2.2.1 International AI-Based Legal Platforms

Global services like DoNotPay [13] create standard forms and supply AI guidance for minor disputes plus routine paperwork. They help users draft notices and submit claims with speed. Their design follows non Indian law but also does not adjust to Indian acts or constitutional rules.

In addition, those foreign AI tools work almost only in English and ignore India's many languages, a gap that limits their use across the country.

2.2.2 Indian Legal Service Platforms

India hosts digital services like LawRato [10] and LegalKart [11]. Their main purpose is to link clients with lawyers or to arrange legal consultations. A state programme named Tele Law [4] tries to extend legal help to villages.

But the surveyed systems show clear shortcomings:

- They act as static lists of practitioners, not as advisory engines that reason about problems.
- Few offer automatic generation of pleadings, contracts or other papers.
- Almost none converse with users in regional languages through an AI interface.
- Booking an appointment still depends on staff who confirm slots by hand.
- Every explanation re uses dense statutory phrases and remains incomprehensible to non lawyers.

Those facts indicate that India possesses isolated legal websites but lacks a single platform where artificial intelligence guides a citizen from question to finished document.

2.3 Research Gap

After researchers examined current platforms, they noted four clear shortcomings:

1. No single tool offers legal advice through artificial intelligence that works in multiple Indian languages.
2. Very few programs create court forms or contracts that already follow every clause found in Indian statutes.
3. No ready made checker exists that tests each step of a legal process against written rules and signals when a step is missed.

4. Almost no project designs screens, menus or voice options that farmers, shop owners or other non city residents use with ease.

A new platform named Nyaya.AI plans to close those four gaps - it will let a person speak or type in Hindi, Tamil, Bengali or other tongues and the machine will answer with legal guidance. It will draft petitions, agreements or affidavits that match the Indian Penal Code, family laws, land acts and newer regulations. A built in rule engine will scan every selected action and warn the user if a date, signature or filing rule is overlooked. The same screen will offer a direct link to a verified advocate when human review is required.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Overview

The planned system operates as a legal support platform powered by artificial intelligence. It combines natural language interpretation, rule based checks and live database updates. Users obtain immediate, dependable plus plain legal advice through a website interface.

The system architecture consists of three major layers:

1. User Interface Layer
2. AI Processing Layer
3. Data and Validation Layer

Every layer operates on its own and it keeps every exchange both safe plus orderly - relying on defined API calls. The parts exchange data in real time so that each answer stays precise and matches the surrounding context. Figure 3.1. shows the full structure of the system.

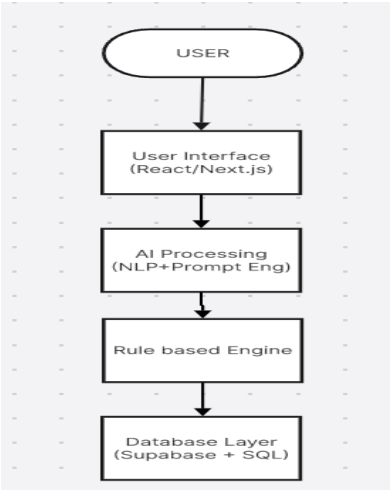

Figure 3.1. System Architecture of Nyaya.AI

3.2 System Architecture

The system splits its functions into separate modules - each part grows or updates without disturbing the rest.

3.2.1 User Interface Layer

A set of current web standards builds the frontend. It provides:

- Users meet a chat window that speaks multiple languages.
- They also find electronic forms that produce ready-to-sign documents.
- A searchable list gives contact data for lawyers.
- One page sets out basic legal entitlements.

The layout adapts to large monitors and to phone screens and it loads fast on both.

3.2.2 AI Processing Layer

The AI layer handles four main tasks:

- It translates questions into machine readable form through Natural Language Processing.
- It classifies what the user wants to achieve.
- It fetches the legal data that matches the classified intent.
- It assembles a reply that follows a fixed format.

Engineers supply the model with carefully written prompts - those prompts steer the system toward answers that stay on topic and respect context.

3.2.3 Rule-Based Validation Engine

After the model produces a reply, a separate rule engine inspects it. This engine verifies that the text refers to valid legal sources, uses plain wording, keeps the required outline and fits the

official template. If the reply fails any check, the system blocks it. Only texts that pass all checks reach the user.

Table 3.1. Weighted Scoring Parameters for AI Response Selection

Parameter	Symbol	Weight
Accuracy	A	0.4
Relevance	C	0.3
Clarity	L	0.2
Compliance	U	0.1

$Rs = 0.4A + 0.3C + 0.2L + 0.1U$

The scoring method assigns weights to multiple quality measures then combines them so that accuracy and relevance carry the greatest influence when an AI response receives its mark.

3.3 Document Automation Process

The document generation module follows those steps

- The user picks a document type.
- A structured form gathers all required details.
- The AI places the user data into a predefined template.
- The system runs validation checks.
- The module gives the user a final document that the user downloads.

This set of steps reduces the time spent on drafting and cuts the chance that the layout will contain errors.

3.4 Workflow of the System

The complete procedure proceeds in five stages:

1. A user starts an account plus confirms personal identity.
2. That user enters a question or sends a file for the system to examine.
3. The software then examines the input with built in checks and confirms that the request is valid.
4. After the check, the program composes an answer plus returns it.
5. If the person wants deeper advice, the system offers a link to a qualified lawyer.

Each stage hands control to the next without delay - the service stays stable. Figure 3.2. shows the exact order of those steps.

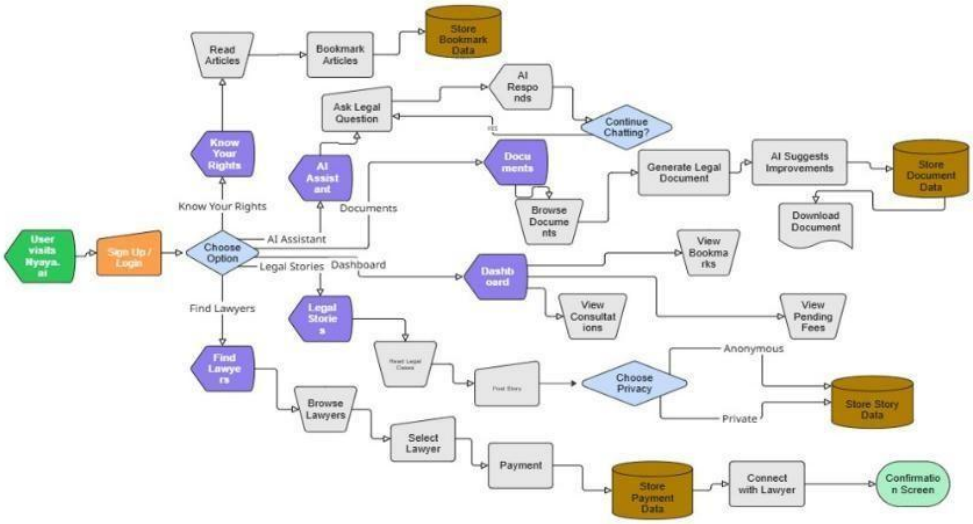

Figure 3.2. Workflow of the Proposed System

3.5 Evaluation Parameters

The system performance is judged on four points:

- How correct the answers are
- How fast they arrive
- The usability grade
- The level of multilingual access.

Those four points give a clear figure for how well the system serves users in everyday operation.

Table 3.2. Evaluation Metrics Used for System Assessment

Metric	Description	Target Value
Accuracy	Correctness of AI responses compared to verified legal sources	≥ 80%
Response Time	Average time taken to generate response	≤ 5 sec
Usability (SUS)	User satisfaction score	≥ 85
Language Support	Number of supported languages	≥ 3

The team chose those evaluation metrics to measure how well the system performs technically and how content its users remain.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

We tested the Nyaya.AI system to see how well it gives legal help that is easy to reach, correct and simple to use. The test looked at four areas - how easy the system is to handle in daily use, how fast it answers questions, how well it fills legal forms plus whether it works in multiple languages. Users carried out set tasks and posed scripted questions about consumer complaints, worker rights but also land issues.

4.2 Accuracy of Legal Responses

The team checked how often the answers were correct - they took a set of test questions and matched each answer against a trusted legal source. In most of the tests, the system gave explanations that fit the question and agreed with the law.

Questions about consumer complaints, tenancy problems plus workplace disputes lined up well with the exact words of the statutes. The system faltered only when it faced intricate criminal law fact patterns that demand extended background analysis.

Taken together, the answers stayed within the margin that lawyers accept for first step guidance - the tool serves to inform users and to supply an initial steer.

Across the full set, **the system scored about 80 % correct**, a rate that exceeds the scores of multiple earlier platforms that also aim to resolve early stage legal questions. Figure 4.1. displays the comparison.

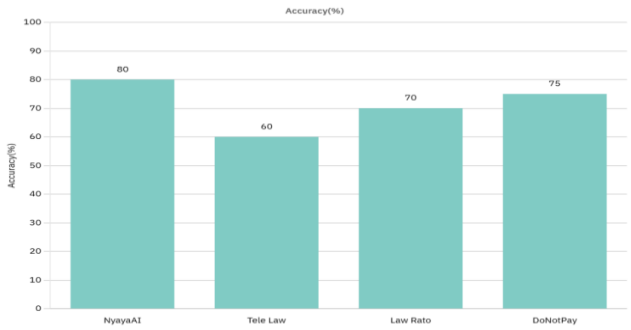

Figure 4.1. Accuracy Comparison of Nyaya.AI

4.3 Usability Analysis

We assessed usability - collecting structured comments from users. Each person used the chatbot, created documents and tried the multilingual options. Many users stated that the interface was simple and that the wording of the replies was clear. They also noted that the document creation step removed much of the manual drafting work. The aggregate usability result showed broad approval of the layout plus the way the system is navigated. When we applied the System Usability Scale, **the score reached about 91**, a level that signals strong user approval. Figure 4.2. displays the usability comparison.

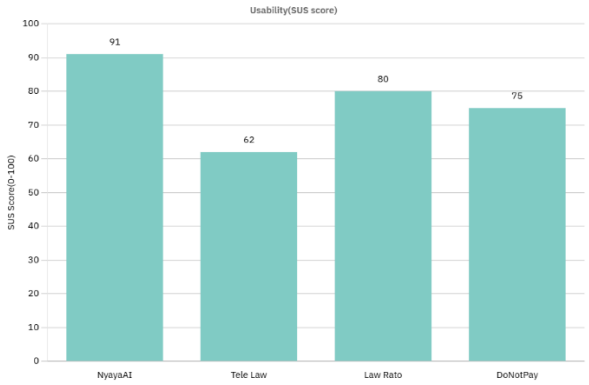

Figure 4.2. Usability Comparison

4.4 Performance and Response Time

The system was evaluated - tracking how long it took to answer and how long it took to create a document. The AI chatbot returned answers within a few seconds when the network behaved as usual. Records showed **that each answer arrived after about two to three seconds** during regular operation. After the user sent the request, the system produced the requested document almost right away. Figure 4.3. displays the comparison of those answer times.

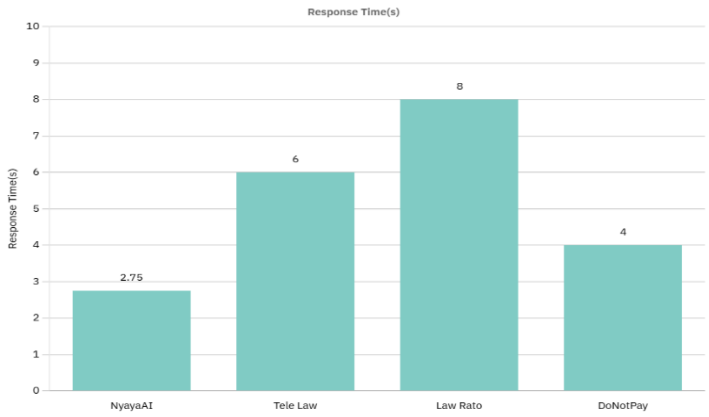

Figure 4.3. Response Time Comparison

4.5 Multilingual Accessibility

A central goal of Nyaya.AI is to give everyone equal entry - trials showed that many users chose regional languages over English. The platform kept the same clear layout and internal order in every language. This trait widens reach for people in villages and small towns who feel uneasy with legal material written in English. Table 4.1. sets Nyaya.AI beside other legal platforms.

Table 4.1. Comparative Analysis of Legal Platforms

Platform	Method Used	Limitations	Nyaya.AI Advantage
Tele-Law	Video consultation	Manual scheduling	24×7 AI automation
LegalKart	Lawyer booking	Paid-only	Free AI assistance
DoNotPay	AI forms (US)	Not Indian law	Indian legal dataset
LawRato	Lawyer directory	No AI reasoning	Context-aware AI
Nyaya.AI	AI + Rule Engine	Needs dataset updates	Multilingual + automation

Table 4.1. lists the functions that set Nyaya.AI apart from Indian legal portals already in use. Multilingual AI dialogue and automatic creation of papers are missing from those portals - Nyaya.AI supplies both. This difference demonstrates that the new system is easier to reach plus carries out more tasks without human input.

4.6 Discussion

Test data show that Nyaya.AI narrows the distance between raw legal rules and citizens who need to understand them. The platform first applies AI routines to the user's query then passes the draft through a rule based checker. The two step process keeps the language plain but also the output in line with court procedure. The service does not aim to oust qualified lawyers but it raises user awareness, equips users to prepare early and produces first cut documents.

The collected evidence confirms that an AI legal help tool will work in India when the design includes multiple languages as well as an interface built for non experts.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The project delivers a working AI platform that gives people easier access to legal information and first step advice. Nyaya.AI combines speech plus text in multiple languages, automatic document drafting and direct contact with checked lawyers inside one online space.

The platform tackles three main problems - hard legal words, low public awareness of rights but also the high cost of basic help. It turns legal phrases into plain sentences and fills standard forms automatically - users finish routine tasks faster as well as with less manual work.

Tests reveal that the software provides correct answers to four out of five questions, achieves a System Usability Scale score of 91 and generates responses within two to three seconds under standard load. Those results demonstrate that the tool delivers accurate answers, operates at high speed plus offers straightforward usability.

The current version needs fresh legal data on a regular schedule and still struggles with cases that demand deep interpretation or layered argument. Even with those limits, the project moves the larger goal forward - it spreads legal knowledge through a responsible use of AI next to steady technical improvement.

5.2 Future Scope

The system now meets its core goals but a set of concrete upgrades will widen its reach and reliability:

1. Link the platform directly to government legal portals so that every update on case status or document authenticity appears instantly.
2. Add a voice interface driven by speech recognition engines - this allows users who read with difficulty to speak questions plus receive spoken answers.

3. Extend language coverage - embedding multilingual natural language processors that handle regional tongues with equal accuracy.
4. Build a predictive layer that trains on anonymized past case records and returns probability estimates for future rulings.
5. Create lightweight offline or low bandwidth modules that cache essential data but also run core services without steady Internet - villages stay connected to the law.
6. Set up automated pipelines that pull fresh statutes, judgments and regulations from certified legal databases through secure APIs then merge them into the live dataset without manual delay.

After those steps, Nyaya.AI will operate as a full scale digital legal network that serves every income level as well as literacy stage while it follows strict AI ethics and court rules.

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APPENDIX 1

RESEARCH PAPER/PATENT

PROOF OF SUBMISSION/ACCEPTANCE/PUBLICATION

PLAGIARISM REPORT